Lab 10.2 R Setup Instructions

Step 1: Install R

Mac: Go to https://cran.r-project.org/bin/macosx/big-sur-arm64/base/ and download the latest version (4.5.3). Choose the installer that matches your chip (Apple Silicon for M1/M2/M3/M4, Intel for older Macs). Open the downloaded file and follow the prompts.

Windows: Go to https://cran.r-project.org/bin/windows/base/old/ and download version 4.5.3. Run the installer and accept the defaults.

Step 2: Install RStudio

Go to https://posit.co/downloads and download **RStudio Desktop** (free). Install R first, then RStudio. Open RStudio afterward to confirm it launches.

Step 3: Download the lab folder

Download the folder "**Sci_Comp26"** from the Box link and save it somewhere easy to find, like your Desktop. Note the full path to the folder, because you'll need it in the next step.

Step 4: Install the packages

Open RStudio and run the following in the **Console**. You only need to do this once.

```
# CRAN packages

install.packages(c("terra", "rgbif", "robis", "RJSONIO", "ecospat",

                   "blockCV", "xgboost", "devtools", "sf", "ggplot2"))


# Bio-ORACLE package from GitHub

devtools::install_github("bio-oracle/biooracler")



# SDM package from the lab folder: replace the path with where YOUR Sci_Comp26 folder is

install.packages("/path/to/Sci_Comp26/bDSSDMTools_0.0.0.9000.tar.gz",

                 repos = NULL, type = "source")
```

Tip: to get the file path, find bDSSDMTools_0.0.0.9000.tar.gz in your folder. On Mac, right-click it, hold Option, and choose "Copy as Pathname." On Windows, hold Shift, right-click, and choose "Copy as path" (then change \ to / and remove the quotation marks if needed.

Step 5: Load the packages

Run this at the start of every R session:

```
library(terra)

library(bDSSDMTools)

library(ggplot2)

library(sf)
```

If this runs without errors, you're ready for the lab.

Troubleshooting

- **Installation errors on Mac:** Some packages need Apple's command line tools. Run xcode-select --install in the Terminal, then try again.
- **Installation errors on Windows:** Some packages need Rtools. Install the version matching your R version from https://cran.r-project.org/bin/windows/Rtools/.
- **"Package not found" or "there is no package called...":** Re-run the install line for that package and check for typos.
- **If asked "Do you want to install from sources?":** Type no (binary versions install faster and with fewer problems).
- **File paths in the lab script** (such as /Users/new/Desktop/Sci_Comp26/...) must be changed to where the folder is on your computer.